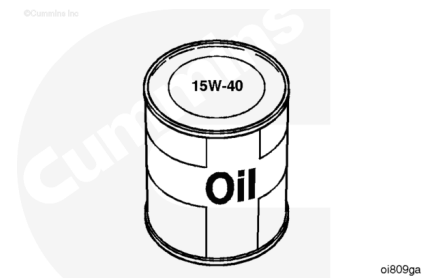


Trainings ▾

General Information

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid the possibility of inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

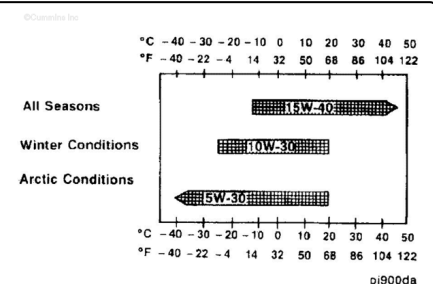


⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

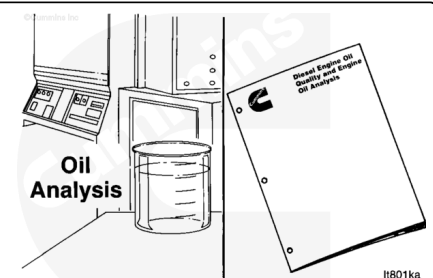
⚠ CAUTION ⚠

The use of low viscosity oil, such as 10W or 10W-30, can be used to aid in starting the engine and in providing sufficient oil flow at ambient temperatures below -5°C [23°F]. Continuous use of low viscosity oil can decrease engine life due to wear.



A used oil analysis can help diagnose internal damage and determine if it was caused by one of the following:

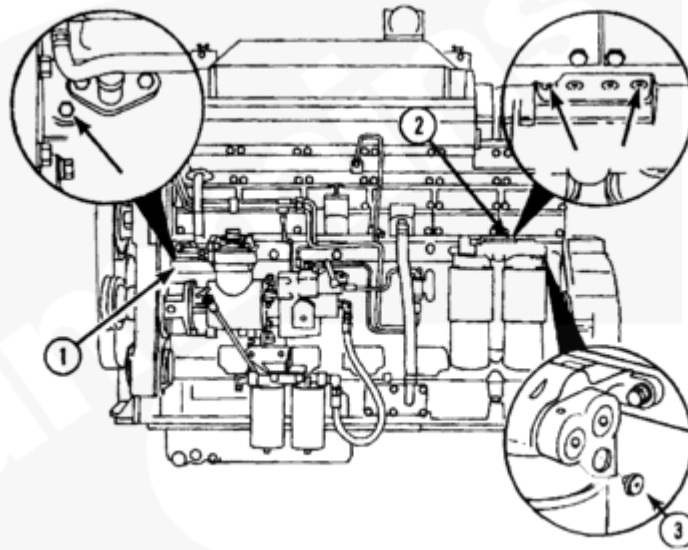
- Intake air filter malfunction
- Coolant leaks
- Oil diluted with fuel
- Metal particles causing wear.



Do **not** disassemble an engine for repair based **only** on the results of an oil analysis. Also, inspect the oil filter. If the filter shows evidence of internal damage, find the source of the

problem and repair the damage. Refer to Procedure 007-002 in Section (/qs3/pubsys2/xml/en/procedures/20/20-007-002-om.html) for oil filter inspection.

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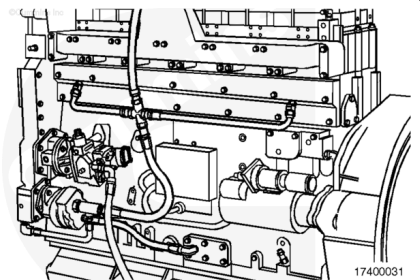
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Oil Pressure Ports

1. Engine oil rifle pressure plug
2. Plug - filtered oil
3. Plug - unfiltered oil.

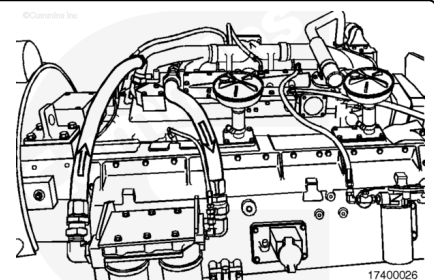
Rail Applications

Because the rail engine is horizontally mounted, a gear driven lubricating oil scavenge pump is used to scavenge lubricating oil collected in the front gear cover, cam follower cover and from the turbocharger lubricating oil drain line. Lubricating oil is scavenged from these three locations and pumped to a hand hole cover in the side of the lubricating oil pan adapter. The scavenge pump is gear driven off the front hydraulic drive. The scavenge pump has been developed for improved low speed capability, better overall efficiency, enhanced flexibility and improved reliability. The all cast iron construction adds to pump durability and noise reduction.



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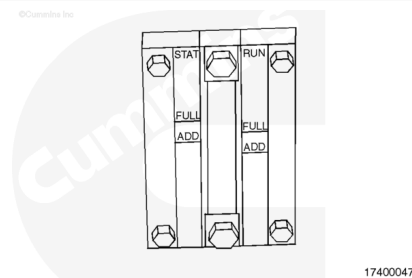
The lubricating oil filter head is remote mounted on the lubricating oil pan by a bracket. Lubricating oil supply to the filters is taken from the lubricating oil cooler housing. The filtered lubricating oil is then returned to an oil transfer connection on the lubricating oil cooler housing.



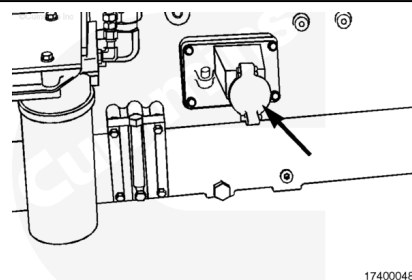
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Note : The QSK19 rail engine does not have a dipstick.

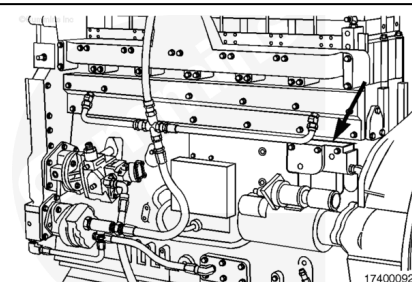
Lubricating oil level can be checked by the use of a sight glass mounted on the lubricating oil pan. The sight glass has marks for checking both HIGH (FULL) 64 liters [17 gal] and LOW (ADD) 53 liters [14 gal] lubricating oil levels at either idle speed (RUN) or while the engine is shut off (STATIC).



The lubricating oil fill is located on the side of the lubricating oil pan.



The lubricating oil transfer connection is mounted on the intake side of the engine.



Filtered lubricating oil enters the lubricating oil transfer connection from the lubricating oil filter head (B) and passes through the lubricating oil transfer connection to the main oil rifle (A) and piston cooling nozzle control valve (D) and to the piston cooling nozzle rifle (C).

